

ReelMind

Database Schema — Full Specification

7 tables · Supabase (PostgreSQL + pgvector) · All tables use UUID primary keys · RLS enabled on all tables

KEY	PK	FK	UK	IDX
Meaning	Primary Key	Foreign Key	Unique Constraint	Indexed Column

profiles			schema: public
Extends Supabase auth.users. Created automatically on signup via a trigger. One row per user.			
Column	Type	Key	Description
id	UUID	PK	FK → auth.users.id. Primary identifier for all user-owned data.
created_at	TIMESTAMPTZ	IDX	Auto-set on insert.
updated_at	TIMESTAMPTZ		Auto-updated via trigger on any row change.
Indexes	id (PK, auto)		

categoriesschema: public

Stores both the 6 seeded default categories (*user_id* = NULL, *is_default* = TRUE) and any user-created ones. Query: WHERE *user_id* = \$uid OR *is_default* = TRUE.

Column	Type	Key	Description
id	UUID	PK	gen_random_uuid(). Primary key.
user_id	UUID	FK	FK → profiles.id. NULL for the 6 default system categories. NOT NULL for user-created ones.
name	TEXT	UK	Unique per user. Unique constraint on (user_id, name). Case-insensitive collation recommended.
is_default	BOOLEAN		TRUE for the 6 seeded categories. FALSE for all user-created ones. Defaults to FALSE.
created_at	TIMESTAMPTZ		Auto-set on insert.
Indexes	(user_id, name) UNIQUE — prevents duplicate category names per user		
Relations	user_id → profiles.id (SET NULL on delete — default categories survive user deletion)		

reels			schema: public
Core table. One row per saved reel per user. Soft-deleted via deleted_at. Hard deletion of reel_chunks happens immediately on soft delete.			
Column	Type	Key	Description
id	UUID	PK	gen_random_uuid().
user_id	UUID	FK	FK → profiles.id. CASCADE DELETE — all reels removed when user account deleted.
category_id	UUID	FK	FK → categories.id. NULL when reel is Uncategorized. SET NULL on category delete.
url	TEXT	UK	Original Instagram reel URL. Unique per user — (user_id, url) constraint prevents duplicate saves.
creator_handle	TEXT		Instagram handle of the creator. Extracted at ingestion time. Nullable.
thumbnail_url	TEXT		URL pointing to cached thumbnail in Supabase Storage. Nullable until processing completes.
transcript	TEXT		Full Whisper transcript. Nullable — empty for music-only reels with no caption fallback.
caption	TEXT		Reel caption text extracted at ingestion. Nullable.
hashtags	TEXT[]		Array of hashtags extracted from caption. Defaults to empty array {}.
summary	TEXT		2-line summary generated by LLM during classification. Shown on library cards.
confidence	FLOAT4		LLM classification confidence score 0.0–1.0. NULL until classification runs.
status	TEXT	IDX	Enum-like: 'queued' 'processing' 'ready' 'failed' 'uncategorised'. Indexed for dashboard queries.
retry_count	SMALLINT		Number of transcription/processing retries attempted. Defaults to 0. Max 3.
deleted_at	TIMESTAMPTZ	IDX	Soft delete timestamp. NULL = active. Non-null = deleted. Indexed for RLS filtering.
created_at	TIMESTAMPTZ	IDX	Save timestamp shown on library cards. Indexed for ordering.
updated_at	TIMESTAMPTZ		Auto-updated via trigger.
Indexes	(user_id, url) UNIQUE — prevents duplicate saves · (user_id, category_id) — fast category library loads · status — filter by processing state · deleted_at — RLS soft-delete filtering · created_at DESC — ordered library view		
Relations	user_id → profiles.id (CASCADE DELETE) · category_id → categories.id (SET NULL on delete)		

reel_chunks			schema: public
Each reel transcript is split into ~200-token chunks. Each chunk gets its own embedding vector. Hard-deleted immediately when parent reel is soft-deleted — removes it from the RAG pipeline.			
Column	Type	Key	Description
id	UUID	PK	gen_random_uuid().
reel_id	UUID	FK	FK → reels.id. CASCADE DELETE — chunks are wiped when reel is deleted (hard or soft).
user_id	UUID	FK	FK → profiles.id. Denormalised here to allow efficient per-user pgvector similarity search without joining reels.
chunk_index	SMALLINT		Zero-based position of this chunk in the full transcript. Used to reconstruct order for context windows.
content	TEXT		The raw text of this chunk (~200 tokens). Sent to LLM as retrieved context.
embedding	vector(1536)	IDX	OpenAI text-embedding-3-small output. 1536 dimensions. Indexed with ivfflat for ANN search.
created_at	TIMESTAMPTZ		Auto-set on insert.
Indexes	embedding vector_cosine_ops WITH ivfflat — ANN similarity search · (user_id, reel_id) — filter chunks by user and reel		
Relations	reel_id → reels.id (CASCADE DELETE — hard deletes chunks on reel deletion) · user_id → profiles.id (CASCADE DELETE)		

chat_sessions			schema: public
One session per conversation thread. category_id = NULL means global search (cross-category). Sessions persist across app restarts so users can review old chats.			
Column	Type	Key	Description
id	UUID	PK	gen_random_uuid().
user_id	UUID	FK	FK → profiles.id. CASCADE DELETE.
category_id	UUID	FK	FK → categories.id. NULL for global search sessions. SET NULL if category is deleted.
title	TEXT		Auto-generated from the first user message (truncated to 60 chars). Shown in session history list.
created_at	TIMESTAMPTZ	IDX	Session start time. Used to sort history newest-first.
updated_at	TIMESTAMPTZ	IDX	Updated on every new message. Used to surface recently active sessions.
Indexes	(user_id, updated_at DESC) — ordered session history list · (user_id, category_id) — filter sessions by category		
Relations	user_id → profiles.id (CASCADE DELETE) · category_id → categories.id (SET NULL on delete)		

chat_messages

schema: public

One row per message turn. role = 'user' or 'assistant'. sources JSONB stores which reels/web results backed the assistant's response — used to render rich cards.

Column	Type	Key	Description
id	UUID	PK	gen_random_uuid().
session_id	UUID	FK	FK → chat_sessions.id. CASCADE DELETE — messages removed with session.
role	TEXT		'user' or 'assistant'. CHECK constraint enforces these two values only.
content	TEXT		Full message text. For assistant messages this is the LLM-synthesised response.
sources	JSONB		Array of source objects: [{type: 'library' 'web', reel_id, chunk_id, creator_handle, thumbnail_url, extract, url}]. NULL for user messages.
created_at	TIMESTAMPTZ	IDX	Message timestamp. Used to render conversation in order.
Indexes	(session_id, created_at ASC) — ordered message fetch per session		
Relations	session_id → chat_sessions.id (CASCADE DELETE)		

feedback			schema: public
<i>Thumbs up (+1) / thumbs down (-1) on assistant responses. One row per user per message — unique constraint prevents double-voting. Used for PRD Section 8 quality metrics.</i>			
Column	Type	Key	Description
id	UUID	PK	gen_random_uuid().
user_id	UUID	FK	FK → profiles.id. CASCADE DELETE.
message_id	UUID	FK	FK → chat_messages.id. CASCADE DELETE.
rating	SMALLINT	UK	+1 (thumbs up) or -1 (thumbs down). CHECK constraint enforces only these two values. Unique on (user_id, message_id).
created_at	TIMESTAMPTZ		Auto-set on insert.
Indexes	(user_id, message_id) UNIQUE — one vote per user per message · message_id — aggregate ratings per message		
Relations	user_id → profiles.id (CASCADE DELETE) · message_id → chat_messages.id (CASCADE DELETE)		

From	Card.	To	Card.	Via FK	Description
profiles	1	categories	0..*	user_id	A user creates zero or more custom categories
profiles	1	reels	0..*	user_id	A user saves zero or more reels
profiles	1	chat_sessions	0..*	user_id	A user starts zero or more chat sessions
profiles	1	feedback	0..*	user_id	A user gives zero or more feedback ratings
categories	1	reels	0..*	category_id	A category contains zero or more reels (NULL = Uncategorised)
categories	1	chat_sessions	0..*	category_id	A category scopes zero or more sessions (NULL = global)
reels	1	reel_chunks	1..*	reel_id	A reel is split into one or more chunks for embedding
chat_sessions	1	chat_messages	1..*	session_id	A session contains one or more messages
chat_messages	0..1	feedback	0..1	message_id	An assistant message may receive one feedback rating per user

Edge Case: Deleted Reel Referenced in Chat History

When a reel is deleted (deleted_at set), its reel_chunks rows are hard-deleted immediately, removing it from the RAG pipeline entirely. However the reel row itself is soft-deleted so it remains referenceable. The sources JSONB on existing chat_messages may contain the reel_id of this now-deleted reel. The iOS app must handle this gracefully: when rendering a source card, check whether the referenced reel has deleted_at set. If so, render a degraded card showing "This reel was deleted" instead of the thumbnail and Watch Reel button. The assistant text response is unaffected — only the source card degrades. A full solution (e.g. flagging affected messages, notifying the user) is deferred to a later pipeline as noted in the PRD.